DRG. NO. ARVR-001 · SHEET 1 / 4

AR/VR LAB

HIRACHAND DAULAT LABORATORY · COEP TECH

SECTION A · THE HARDWARE

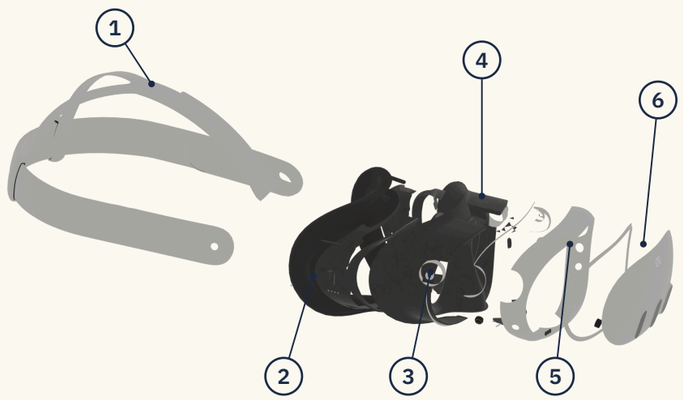
Four headsets, taken apart

Every XR headset in the lab, drawn from real 3D product models and exploded into its main parts. Numbers on each drawing match the list beside it.

Explore it in 3D

Scan for the interactive web brochure.

FIG. 1A · META QUEST 3



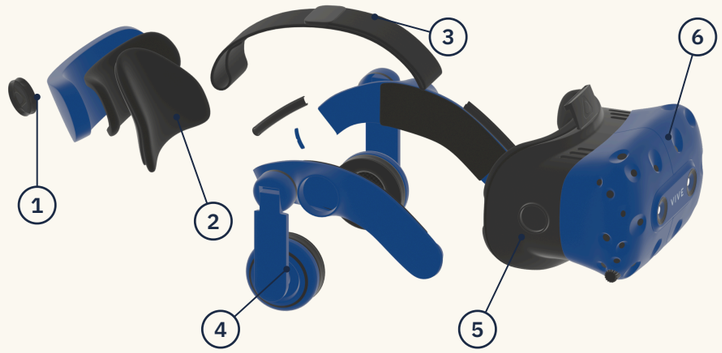
Meta Quest 3

VR · MR · Standalone, also PC VR

DISPLAY	Dual LCD, 2064 × 2208 per eye
FOV	110°
TRACKING	Inside-out 6DoF, hand tracking
CHIP	Snapdragon XR2 Gen 2
WEIGHT	500 g · 2–3 h battery

1. **Head strap** · holds 500 g steady for long sessions
2. **Facial interface** · cushions the face, seals out light
3. **Pancake lenses + LCDs** · 2064 × 2208 per eye, 110° view
4. **Chassis + XR2 Gen 2** · all compute on the headset
5. **Housing** · battery for 2–3 h untethered
6. **Front cover + sensor pills** · colour passthrough, depth, hand tracking

FIG. 1B · HTC VIVE PRO 2



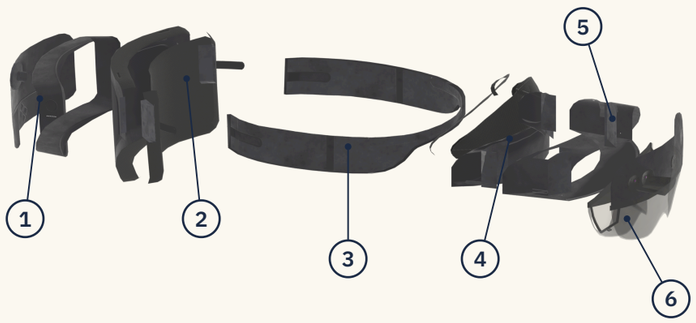
HTC Vive Pro 2

VR · PC-tethered

DISPLAY	RGB LCD, 2448 × 2448 per eye (5K)
FOV	120°
REFRESH	90 / 120 Hz
TRACKING	Lighthouse base stations, 6DoF
WEIGHT	850 g

1. **Rear pad + fit dial** · tightens the fit
2. **Rear cradle** · spreads the load over the head
3. **Top strap** · takes weight off the face
4. **Built-in headphones** · positional audio
5. **Body + face cushion** · displays and lenses, cable to the i9 PC
6. **Front shell** · Lighthouse sensors + stereo cameras

FIG. 1C · MICROSOFT HOLOLENS 2



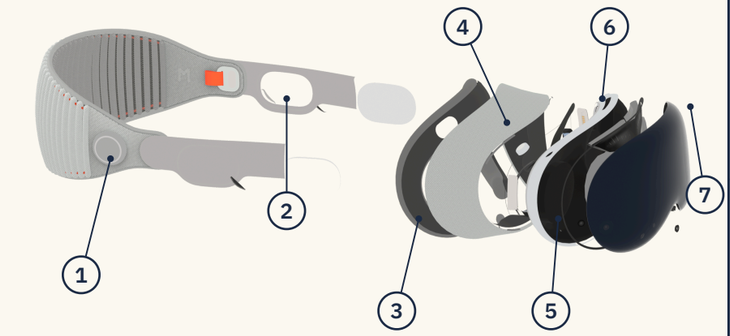
Microsoft HoloLens 2

AR · Standalone holographic

DISPLAY	2K 3:2 light engines, 2048 × 1080 per eye
FOV	About 52°
CHIP	Snapdragon 850
INPUT	Hands, eyes, voice; no controllers
WEIGHT	About 566 g

1. **Battery pack + dial** · balances the headset at the back
2. **Rear housing** · battery and electronics
3. **Headband** · holds the display in front of the eyes
4. **Brow pad + compute** · Snapdragon 850 runs apps
5. **Sensor bar + light engines** · 6DoF, hand and eye tracking; projects holograms
6. **Visor + waveguides** · see-through holograms, about 52°

FIG. 1D · APPLE VISION PRO



Apple Vision Pro

MR · Mixed reality, AR + VR

DISPLAY	Micro-OLED, ≈ 4K per eye est.
FOV	About 120° est.
CHIPS	Apple M2 + R1
INPUT	Hands, eyes, voice
WEIGHT	500–600 g est.

1. **Solo Knit Band + Fit Dial** · spreads the weight
2. **Audio straps** · spatial audio near the ears
3. **Cushion** · soft layer against the face
4. **Light seal** · blocks outside light
5. **Lens + display modules** · micro-OLED, ≈ 4K per eye
6. **Aluminium frame, M2 + R1** · apps and sensor processing
7. **Glass front** · covers the camera and sensor array

Compared to scale

Hatched bars are the maker's unconfirmed figures. Also in the lab: an i9-13900K workstation (64 GB RAM) and an NVIDIA RTX 4090 for model training.



3D models via Sketchfab, CC BY 4.0: Meta Quest 3 by ElinHohler · HTC Vive Pro by EternalRealm (the lab's Vive Pro 2 shares its shell) · HoloLens 2 by muertoHambre · Apple Vision Pro by jnanbr07. Specs from the lab's device list.

TITLE	SHEET	SCALE
FOUR HEADSETS, TAKEN APART	1 / 4	NTS

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AR/VR LAB

HIRACHAND DAULAT LABORATORY · COEP TECH

SECTION B · THE RESEARCH

Detail views: what we are building

Six projects from the lab's own reports, from surgery planning to rider safety. Each one uses AR, VR or the computer vision behind them.

Explore it in 3D

Scan for the interactive web brochure.



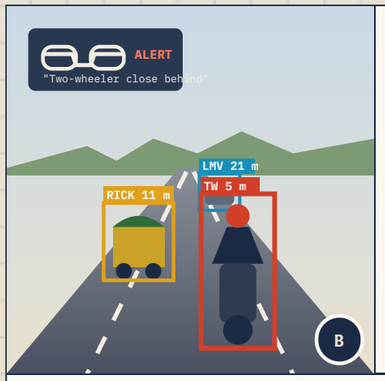
ROBOTICS · META QUEST 3

VR control of a collaborative robot

The operator moves a virtual copy of the lab's cobot with Quest 3 controllers, and the real arm follows.

- Controller motion is sent to the robot as position deltas (DELTA).
- Two buttons send GRIP_OPEN and GRIP_CLOSE to the gripper.
- The robot reports CUR_POS back, so the virtual and real arms never drift.
- Clearer sense of the arm's path than voice or gesture control; easier to learn than a joystick.

Viraj Bakshi, Prasad Patil, Amit Deshmukh, Pratiksha Jawale (research scholar) · **Guide:** Dr. Shantipal S. Ohol



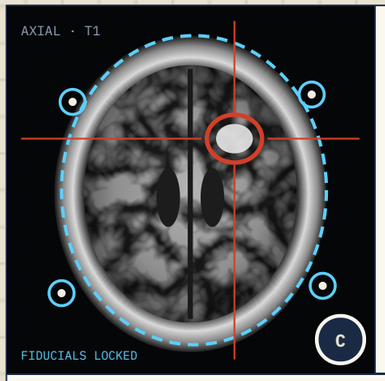
WEARABLE ADAS · ROAD SAFETY

AI glasses for smart mobility

Two-wheelers are nearly two-thirds of vehicles on Indian roads. This wearable watches the road and warns the rider.

- Perception: a camera feed and YOLO detection of vehicles, pedestrians and obstacles.
- Intelligence: distance from box size, speed and trajectory; crash detection from IMU data.
- Interface: audio alerts in the AI glasses and an automatic SOS with GPS location.
- Tested on the road: few false alerts and SOS within seconds of a crash.

AR/VR Lab team · Computer vision, embedded AI, IMU and GPS



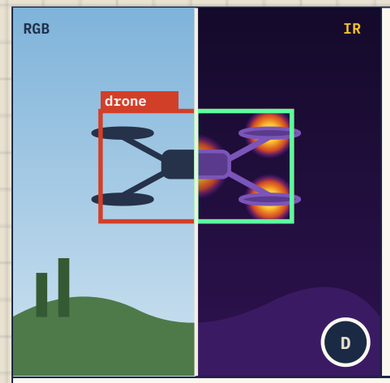
MEDICAL AR · REGISTRATION

MRI overlay for pre-surgical planning

Surgeons plan an approach with the scan placed on the real patient.

- Fiducial markers map virtual coordinates to real ones.
- Point pairs are picked, locked and aligned in an interactive UI.
- Rigid fit with the SVD-based Kabsch algorithm, refined by Iterative Closest Point.
- Registration accuracy is measured for every alignment.

Sharanya R. Dumbre Patil, M.Tech Instrumentation · **Supervisor:** Prof. U. M. Chaskar



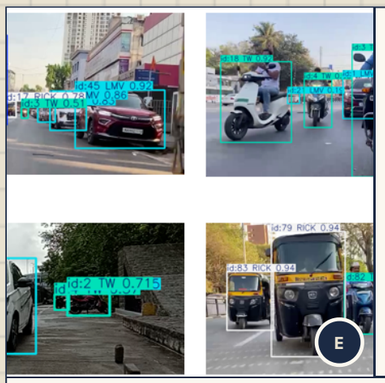
COMPUTER VISION · RGB + INFRARED

Drone identification and threat response

Reliable detection by day and night is the first step to any response.

- XL variants of YOLOv8, v9, v10, v11, v12 and YOLO26 under evaluation.
- Trained on an NVIDIA RTX 4090 with RGB and infrared drone datasets.
- Compared on precision, recall, mAP@50, mAP@50–95, inference time and complexity.

Vaishnavi Taware · **Guide:** Prof. Sunil G. Bhirud



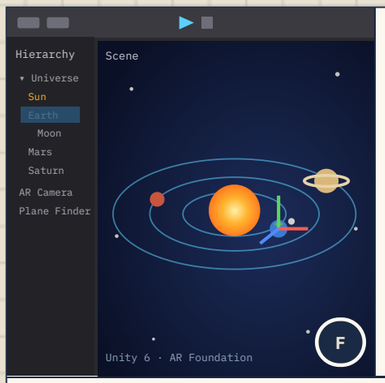
COMPUTER VISION · TWO-WHEELERS

Rear-view critical object detection

A camera watches the road behind the rider, where mirrors can't reach.

- YOLOv8 with tracking on Pune street footage.
- Classes: heavy vehicles (HMV), cars (LMV), auto-rickshaws (RICK), two-wheelers (TW).
- Dataset built for vehicle mix, lighting, augmentation and annotation.
- F1 score of 0.90 across all classes at a confidence of 0.469.

Aditya Rahalkar, B.Tech Mechanical · **Guides:** Dr. S. S. Pardeshi, Dr. Sangeeta Mundra



TEACHING · FINAL-YEAR B.TECH (R&AI)

AR-VR elective with Unity

Students learn HCI, computer vision and graphics by building working AR/VR applications.

- Path: install Unity Hub, basics, an AR project (a 3D universe), samples, scenarios, a working executable.
- Unity 6 with a one-year learning licence; RTX 3050 or better recommended.
- Proposed: project-based evaluation instead of a theory paper.
- Industry link: training and internship talks with Who VR.

Faculty: Dr. Shubhangi Abhijit Deshmukh, Assistant Professor, E&TC

Project details from the AR/VR Lab's project reports. The cobot scene and the street detections are the teams' own images; the other four pictures are illustrations.

TITLE	SHEET	SCALE
DETAIL VIEWS: WHAT WE ARE BUILDING	2 / 4	NTS

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AR/VR LAB

HIRACHAND DAULAT LABORATORY · COEP TECH

SECTION C · THE LAB

Why AR/VR, and where we are going

AR overlays digital content on the real world; VR places you inside a simulated one. Here is why that matters to engineers, and the lab's plan for the next five years.

Explore it in 3D

Scan for the interactive web brochure.

C1 Why AR/VR in engineering?

Design before you build

Walk around a full-scale 3D prototype and catch clashes and errors **before metal is cut**, saving time and cost.

Train without the risk

Practise surgery, machine operation or assembly in simulation, as many times as needed, **with nothing to break**.

See what is invisible

Overlay scan data, stress or flow fields onto real objects, **as in our MRI overlay project**.

Control machines at a distance

Teleoperate robots with natural hand movement, **as the lab does with its cobot**.

Guide hands on the shop floor

Step-by-step AR instructions for assembly and maintenance **reduce errors and training time**.

Make people safer

Perception and timely alerts protect riders and workers, **as in our AI glasses for two-wheelers**.

WHERE IT IS ALREADY USED

- Healthcare and surgical training
- Education
- Architecture
- Engineering design
- Manufacturing and assembly lines
- Automotive
- Corporate training
- Real estate walkthroughs

Global hubs: Silicon Valley, Seattle, Los Angeles, Tokyo, Seoul, Shenzhen, London, Berlin, Tel Aviv. In India, Digital India and Startup India are growing the ecosystem, while cost, awareness and a shortage of skills hold adoption back. Building those skills is this lab's job.

C2 Revision table: the five-year plan

TODAY · SEP 2026

2024	2025	2026	2027	2028
REV A	REV B	REV C	REV D	REV E
Set up the CoE and research framework	Webinars and trainings	PG and professional certification	International technology transfer	State-of-the-art industry hub for AR/VR

FIVE STEPS TO A CENTRE OF EXCELLENCE

01 Research centre
02 Knowledge transfer centre
03 Industry hub
04 Notable IP contribution
05 Internationally recognised CoE

FIVE-POINT INITIATIVE PROGRAMMES

01 Research
Key research titles for IP and publications

02 Learning and development
Webinars, courses, FDPs and ToT trainings

03 Consultancy
Helping businesses adopt AR/VR

04 Technology transfer
Partnerships with international institutes

05 International conference
Hosting a technical conference on AR/VR

OBJECTIVES

- Innovation and research
- Skill development and training
- Best practices and standards
- Collaboration and ecosystem building
- Industry solutions, prototypes and PoCs
- Market awareness and advocacy

C3 In memory



14.04.1932 – 11.05.2021

Shri Hirachand Daulat

A 1955 Mechanical Engineering graduate of COEP. His children, **Dr. Jaldeep and Neeta Daulat**, founded this laboratory in his memory, in the department where he studied.

Supported by Shri Mangesh Gandhi, Mechanical Engineering, 1979.

4 XR HEADSETS: VR, AR, MR	**6** RESEARCH AND TEACHING PROJECTS
RTX 4090 GPU FOR MODEL TRAINING	**2028** TARGET: INDUSTRY HUB

Visit the lab

Dept. of Mechanical Engineering, COEP Technological University, Shivajinagar, Pune
Room [___] · Hours [___] · Contact [___]

Plan, programmes, objectives and context from the lab's presentation (Mechanical Engineering Department, COEP Technological University).

TITLE	SHEET	SCALE
WHY AR/VR, AND WHERE WE ARE GOING	3 / 4	NTS

DRG. NO. ARVR-001 · SHEET 4 / 4

SCAN · STEP INSIDE

HIRACHAND DAULAT AR/VR LAB · COEP TECH, PUNE

Two codes, two ways in. Open your phone camera and point it at either code. Nothing to install.

AUGMENTED REALITY

AR

CODE 1 · AR

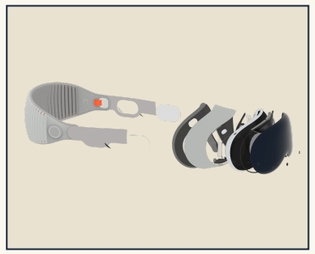
The lab, in your room

Place any of the lab's four headsets on a real table at life size, then explode it into its parts and walk around it.

01 Scan, then tap **View in your room**.

02 Move the phone over a table or the floor.

03 Walk around it; switch to **Exploded**.



Works on Android phones with Google Play Services for AR, and on iPhone or iPad with iOS 12 or later.

OR TYPE
adityapathare46.github.io/coep-arvr-lab/ar

WEB BROCHURE

WEB

CODE 2 · WEB

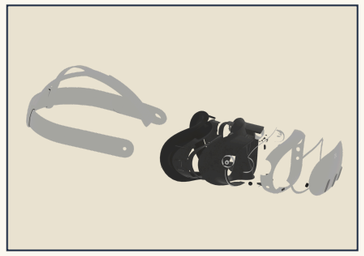
The whole lab, interactive

Take all four headsets apart in 3D, compare their specs, and try working models of the lab's research projects.

01 Scan to open the brochure.

02 Pick a headset and drag the explode slider.

03 Try the cobot, MRI and rider-safety demos.



Works in any modern browser on phones, tablets and laptops.

OR TYPE
adityapathare46.github.io/coep-arvr-lab

AR **Augmented reality**
Digital objects placed in the real world, seen through your phone or a HoloLens 2.

VR **Virtual reality**
A fully simulated world. Try it in the lab on the Meta Quest 3 or HTC Vive Pro 2.

MR **Mixed reality**
Virtual objects that react to the real room, as on the Apple Vision Pro and Quest 3.

Both codes use high error correction and were test-decoded before printing. 3D models via Sketchfab, CC BY 4.0.

TITLE	SHEET	SCALE
SCAN · STEP INSIDE	4 / 4	NTS